

The data Dem data.csv contains 5 variables:

- **district**: the congressional district (1 to 435)
- **Year**: the year of voting (t_{ij} below),
- **Percent Dem**: the percentage of voters that voted Democrat (Y_{ij} below)
- **Percent Dem prev**: the percentage of voters that voted Democrat in the previous election, (X_{ij1} below)
- **Incumbency**: 1 if the incumbent was a Democrat, 0 for open seats, -1 for a Republican incumbent (X_{ij2} below) and a -9 for missing (these need to be recoded as NA's).

The model we model we want to run is:

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + \beta_2 X_{ij1} + \beta_3 X_{ij2} + b_{0i} + \epsilon_{ij} \quad (1)$$

where the variable are defined above.

Missing values correspond to congressional races that were not contested. The dataset contains years 1984, 1986, 1988, and 1990 where, for example, for 1984 the Percent Dem prev variable represents that percentage of democrat voters in the 1982 election.

- a) Qualitatively assess whether the missing data mechanism is MCAR, MAR or MNAR. Be as specific as possible using the vocabulary we discussed in class.
- b) Regardless of your assessment in (a), quantitatively test MCAR vs MAR.
- c) Construct two “bad” imputation procedures and one “good” imputation procedure for the uncontested elections.
- d) Fit a model to each of the imputation methods in (c). Discuss the differences in the results.

(a) If an election is not contested that means there wasn't an opponent who would have been competitive. That is, the election was going to be a landslide (one way or another). As a result, the observations are missing based on their values. Thus, the probability that an observation is missing depends on the value that we don't observe. This is the definition of MNAR.

(b) To qualitatively assess MCAR vs MAR we want to test if the previous value predicts if a value is missing or not. That is, if $R_{ij} = 1$ if a value is missing and 0 otherwise, fit the following logistic model

$$\log\left(\frac{P(R_{ij} = 1)}{1 - P(R_{ij} = 1)}\right) = \beta_0 + \beta_1 Y_{ij-1} + b_{0i}$$

and the hypothesis test $H_0: \beta_1 = 0$ will determine if the previous value is associated with missingness. If this coefficient is significant, we can say that the data is not MCAR since MCAR assumes that the missingness does not depend on the previous value. There will be some Y_{ij-1} values missing, for those we'll have to not use those specific values (but we will use the rest of the observed values for that person). Since we're including a random effect in our model, this will be valid under a MAR assumption.

First, we create the missingness indicator:

```
data Pol_w_miss;
set Political;
Miss_ind = 0;
if Percent_Dem=. then Miss_ind=1;
run;
```

Then we fit the logistic mixed model:

```
proc glimmix data=Pol_w_miss;
class District year;
model Miss_ind = Percent_Dem_prev/solution;
random intercept/sub=District;
run;
quit;
```

Here are the parameter estimates:

Solutions for Fixed Effects

Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	0.03570	0.03003	417	1.19	0.2352
Percent_Dem_prev	0.1973	0.05362	1020	3.68	0.0002

Clearly Y_{ij-1} (here Percent_Dem_prev) is significant indicating the data are not MCAR. This doesn't exactly mean that the data are MAR (they still could be MNAR).

- (c) Construct two “bad” imputation procedures and one “good” imputation procedure for the uncontested elections.

Two bad imputation procedures are mean imputation and last value carried forward. I create both outcomes in the following SAS code (the mean %Dem is 0.525 in this dataset). Note that the last value carried forward doesn't work if the first value is missing. I also set the Incumbency=-9 values to missing (since they are missing, this also helped out the convergence of the models).

```
data Pol_w_miss;
set Political;
Percent_Dem_meanMI = Percent_Dem;
if Percent_Dem_meanMI=. then Percent_Dem_meanMI=0.525;
Percent_Dem_LVCF = Percent_Dem;
if Percent_Dem_LVCF = . then Percent_Dem_LVCF = Percent_Dem_prev;
if Incumbency = -9 then Incumbency = .;
run;
```

Now let's create a multiply imputed dataset. The FCS method implements multivariate imputation by the “chained equations” method to impute values for a data set with an arbitrary missing pattern. It Assumes a joint distribution exists for the data, but does not make any normality assumptions.

```
proc mi data=Pol_w_miss NIMPUTE=10 seed=8675309 out=MI_Politic;
class year Incumbency;
FCS reg (Percent_Dem = year Incumbency Percent_Dem_prev);
FCS reg (Percent_Dem_prev = year Incumbency Percent_Dem);
var Percent_Dem year Incumbency Percent_Dem_prev;
run;
```

- (d) Fit a model to each of the imputation methods in (c).

Now we'll fit the model. The code I used for the first two “bad” procedures:

```
proc mixed data=Pol_w_miss;
class District Incumbency;
model Percent_Dem_meanMI = year Incumbency Percent_Dem_prev;
random /sub=District G;
run;
```

Similarly for the last value carried forward approach. For the MI procedure I used:

```
proc mixed data=MI_Politic;
by _Imputation_;
class District year Incumbency;
```

```

model Percent_Dem = year Incumbency Percent_Dem_prev/S COVB;
random INTERCEPT/sub=District G;
ods output SolutionF=mixparms CovB=mixcovb;
run;

proc mianalyze parms(classvar=full)=mixparms;
class year Incumbency;
modeleffects Intercept year Incumbency Percent_Dem_prev;
run;

```

Here are the parameter estimates for the mean imputation method:

Solution for Fixed Effects

Effect	Year	Incumbency	Estimate	Standard Error	DF	t Value	Pr > t
Intercept			0.2477	0.01017	417	24.36	<.0001
Year	1984		-0.04229	0.006473	1013	-6.53	<.0001
Year	1986		-0.00196	0.006508	1013	-0.30	0.7639
Year	1988		-0.02455	0.006543	1013	-3.75	0.0002
Year	1990		0	.	.	.	.
Incumbency		0	0.07529	0.009969	1013	7.55	<.0001
Incumbency		1	0.1544	0.009263	1013	16.67	<.0001
Incumbency		-1	0	.	.	.	.
Percent_Dem_prev			0.3794	0.02429	1013	15.62	<.0001

And here are the parameter estimates for the LVCF method:

Solution for Fixed Effects

Effect	Year	Incumbency	Estimate	Standard Error	DF	t Value	Pr > t
Intercept			0.1450	0.007771	417	18.65	<.0001
Year	1984		-0.03494	0.004929	1013	-7.09	<.0001
Year	1986		0.01424	0.004955	1013	2.87	0.0041
Year	1988		-0.00912	0.004982	1013	-1.83	0.0674
Year	1990		0	.	.	.	.
Incumbency		0	0.07726	0.007603	1013	10.16	<.0001
Incumbency		1	0.1459	0.007074	1013	20.62	<.0001
Incumbency		-1	0	.	.	.	.

Solution for Fixed Effects

Effect	Year	Incumbency	Estimate	Standard Error	DF	t Value	Pr > t
Percent_Dem_prev			0.5768	0.01855	1013	31.09	<.0001

And here are the parameter estimates for the multiple imputation procedure:

Parameter Estimates (10 Imputations)									
Parameter	year	Incumbency	Estimate	Std Error	95% Confidence Limits		DF	Theta0	t for H0: Parameter=Theta0 Pr > t
Intercept	.	.	0.156994	0.008676	0.13983	0.17416	129.24	0	18.10 <.0001
year	1984.000000	.	-0.040969	0.005880	-0.05268	-0.02926	77.642	0	-6.97 <.0001
year	1986.000000	.	0.014466	0.006365	0.00166	0.02727	47.119	0	2.27 0.0277
year	1988.000000	.	-0.012554	0.005491	-0.02340	-0.00170	149.26	0	-2.29 0.0236
year	1990.000000	.	0	.	.	.	.	0	. .
Incumbency	.	-1.000000	0	.	.	.	.	0	. .
Incumbency	.	0	0.089589	0.008011	0.07386	0.10532	573.2	0	11.18 <.0001
Incumbency	.	1.000000	0.155075	0.007957	0.13933	0.17082	132.04	0	19.49 <.0001
Percent_Dem_prev	.	.	0.547993	0.021532	0.50518	0.59081	84.558	0	25.45 <.0001

We can see that the last value carried forward (LVCF) method performs similarly to the multiple imputation method in terms of the point estimates (e.g., 0.5768 vs 0.5480), but the standard error of the LVCF approach are smaller (e.g., 0.01855 vs 0.02153, which is 16% smaller). The LVCF approach doesn't have much bias since the previous values are unbiased estimates of future values.